جامعة بيرزيت

BIRZEIT UNIVERSITY

Faculty of Engineering and Technology

Electrical and Computer Engineering Department

Artifactual Intelligence

ENCS3340

Project 1

Prepared by:

Mohmmad Omar 1221332

Instructor: Dr. Yazan Abu Farha

Section: 1

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Abstract

This is a system to mimic and optimize the day-to-day activities of a local package delivery store using two metaheuristic algorithms, Simulated Annealing and Genetic Algorithm. The primary objective is to minimize the overall distance traveled by the delivery vans while considering package constraints such as weight, destination, and priority. A package is assigned to one of the limited-capacity vehicles, and Euclidean distance-based optimum routes are computed between points. The system provides dynamic user input for the number of vehicles, vehicle capacities, and package data. Users can select the optimization algorithm dynamically at runtime, and the solution is displayed graphically in a user-friendly interface.

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1. Problem Formulation

1.1 Simulated Annealing problem formulation: -

Initial State	All packages in the shop (0,0) waited to be assigned to a vehicle
Actions (Operators / Successor Functions)	Randomly assign package to vehicle with constrain that the vehicles cannot exceed the capacity.
State Space	All assignments are possible with two constrains (all packages delivered, vehicles don't exceed their capacity).
Goal State	A complete state with all packages delivered with the minimum cost (total distance+ penalty).
Goal Test	Make sure that the goal state occurred.
Path	A sequence of valid package assignment operations (e.g., assign, move, swap) that leads from the initial state to a goal state.
Path Cost	Total Cost=Total Distance+ Priority Penalty
Solution	A state near to the optimal or may be the optimal that minimize the path cost.

1.2 Genetic problem formulation: -

Initial State	A valid random assignment of packages to vehicles where no vehicle exceeds its weight capacity.
Actions (Operators / Successor Functions)	Crossover: Combine two valid vehicle-package assignment plans. Mutation: Move a package from one vehicle to another, if capacity allows (randomly)
State Space	All assignments are possible with two constrains (all packages delivered, vehicles don't exceed their capacity).
Goal State	A complete state with all packages delivered with the minimum cost (total distance+ penalty for high priority).
Goal Test	Make sure that the goal state occurred.
Path	A sequence of genetic operations (initial population → selection → crossover → mutation) that iteratively evolves better solutions over generations.
Path Cost	Total Cost=Total Distance+ Priority Penalty
Solution	A state near to the optimal or may be the optimal that minimize the path cost.

2. Heuristics

2.1 Simulated Annealing heuristics: -

1. **Cost function:** used **g**uides the search toward better solutions.

- Total coat= (Total Distance + priority Penalty)

2. **Neighbor Generation Heuristic:** randomly choose between swap and move to improve the randomly in the distribution.
   - Move: Transfers a package from one truck to another if the destination truck has enough capacity.
   - Swap: Swaps two packages between trucks if they share the same priority and the swap maintains valid capacity constraints.
3. **Stuck state handle:**
   If there is no improvement after a series of temperature cycles, the algorithm performs a random re-distribution of packages to exit local minima and reinsert diversity in the solution space.

2.2 Genetic heuristics: -

1. Fitness Function**:** The algorithm minimizes a custom-defined cost.

- Fitness= − (Total Distance + priority Penalty)
- I choose this syntax to make sure that the lower the cost, the better the solution.

2. **Total Euclidean Distance:** calculate the distance between two points to use it when determine the best solution.

3. **Priority Penalty Heuristic:**

- Additional priority packages are punished to a higher degree if they appear later in the delivery order.
- This stimulates the algorithm to provide critical packages sooner, a domain-specific heuristic.

3. Constraints

3.1 simulated annealing constrains

Constraint	Handle
Vehicle Capacity	Enforced strictly in assignment and neighbor generation
Oversized Packages	Checked and rejected at the beginning via `check_size()` function which handle it
Package Priorities	Handled by adding penalty to the cost in the cost calculation function (not a hard constraint)
Stuck at the same state result	Handled via add counter to check if the same state repeated 10 times then do redistribution

3.2 Genetic constrains

Constraint	Handle
Vehicle Capacity	Enforced strictly in assignment and neighbor generation
Oversized Packages	Checked and rejected at the beginning via `check_size()` function which handle it
Package Priorities	Handled by adding penalty to the cost in the cost calculation function (not a hard constraint)

4. Parameter Used

4.1 Simulated Annealing Parameters:

Parameter	value
Initial Temperature	1000
Cooling Rate	0.99
Stopping Temperature	1
Numbers of iterations per Temperature	100

4.2 Genetic Parameters:

Parameter	value
Population size	50
Mutation rate	0.05
Number of generations	500

5. Parameter Used

Tuning key parameters significantly affect both the performance of Simulated Annealing and the Genetic Algorithm. In Simulated Annealing, the cooling rate controls how gradually the search space is explored lower cooling (e.g., 0.99) allows for greater exploration and solution quality but at a cost of greater runtime. For the Genetic Algorithm, the mutation rate balances exploration and convergence too low produces premature convergence, while too high adds too much randomness. A moderate value (say, 0.05) was seen to be effective in managing diversity without sacrificing the population direction toward optimal solutions. Tuning such parameters appropriately is essential to achieve both efficiency and quality in optimization.

6. Test Cases

Test case -A-

```
(2,5) 15.0 1
(3,1) 14.0 2
(1,6) 12.0 3
(6,6) 11.0 1
(2,4) 13.0 1
(7,7) 10.0 2
(5,1) 12.5 3
(3,5) 13.6 2
(8,2) 14.4 1
(6,3) 12.3 3
```

Figure 2: Test case A packages file

```
30
30
30
30
30
30
30
30
```

Figure 1:Test case B vehicles file

Simulated Annealing output: -

```
Temperature: 0.99, Current Cost: 48.65, Best Cost: 46.10

=== Best Found Assignment ===
Vehicle 0 | Capacity: 30 | Packages: 2
   - Destination: (2.0, 5.0), Weight: 15.0, Priority: 1
   - Destination: (3.0, 5.0), Weight: 13.6, Priority: 2
Vehicle 1 | Capacity: 30 | Packages: 2
   - Destination: (3.0, 1.0), Weight: 14.0, Priority: 2
   - Destination: (5.0, 1.0), Weight: 12.5, Priority: 3
Vehicle 2 | Capacity: 30 | Packages: 0
Vehicle 3 | Capacity: 30 | Packages: 2
   - Destination: (6.0, 6.0), Weight: 11.0, Priority: 1
   - Destination: (7.0, 7.0), Weight: 10.0, Priority: 2
Vehicle 4 | Capacity: 30 | Packages: 0
Vehicle 5 | Capacity: 30 | Packages: 0
Vehicle 6 | Capacity: 30 | Packages: 2
   - Destination: (2.0, 4.0), Weight: 13.0, Priority: 1
   - Destination: (1.0, 6.0), Weight: 12.0, Priority: 3
Vehicle 7 | Capacity: 30 | Packages: 2
   - Destination: (6.0, 3.0), Weight: 12.3, Priority: 3
   - Destination: (8.0, 2.0), Weight: 14.4, Priority: 1
```

Figure 3: Test case A best solution

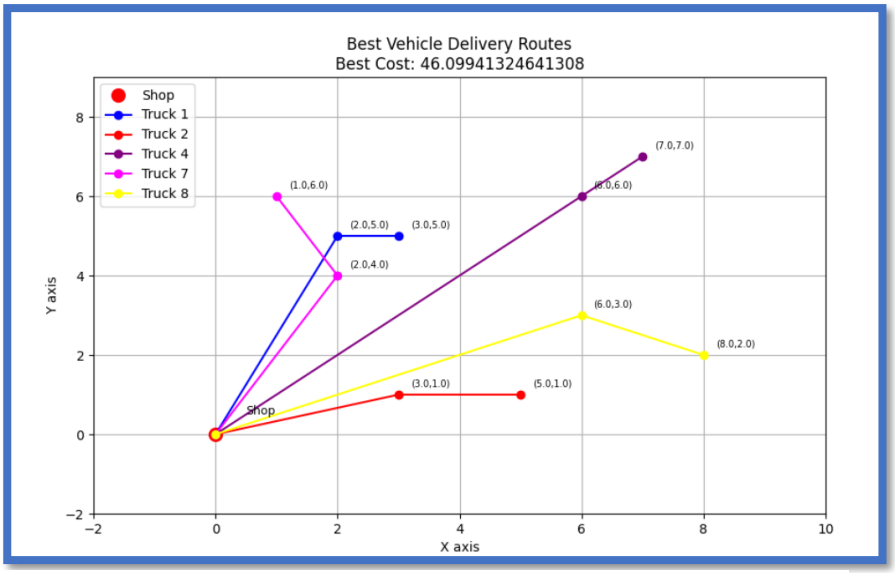


Figure 4: Test case A best solution paths

Genetic output: -

```
Generation 497, Best Fitness: -45.69
Generation 498, Best Fitness: -45.69
Generation 499, Best Fitness: -45.69
Best Total Distance (Cost): 45.69
Vehicle 0 | Capacity: 30 | Packages: 0
Vehicle 1 | Capacity: 30 | Packages: 2
  - Destination: (6.0, 3.0), Weight: 12.3, Priority: 3
  - Destination: (8.0, 2.0), Weight: 14.4, Priority: 1
Vehicle 2 | Capacity: 30 | Packages: 2
  - Destination: (6.0, 6.0), Weight: 11.0, Priority: 1
  - Destination: (7.0, 7.0), Weight: 10.0, Priority: 2
Vehicle 3 | Capacity: 30 | Packages: 2
  - Destination: (3.0, 1.0), Weight: 14.0, Priority: 2
  - Destination: (5.0, 1.0), Weight: 12.5, Priority: 3
Vehicle 4 | Capacity: 30 | Packages: 2
  - Destination: (2.0, 4.0), Weight: 13.0, Priority: 1
  - Destination: (3.0, 5.0), Weight: 13.6, Priority: 2
Vehicle 5 | Capacity: 30 | Packages: 2
  - Destination: (2.0, 5.0), Weight: 15.0, Priority: 1
  - Destination: (1.0, 6.0), Weight: 12.0, Priority: 3
Vehicle 6 | Capacity: 30 | Packages: 0
Vehicle 7 | Capacity: 30 | Packages: 0
```

Figure 6: Test case A best solution

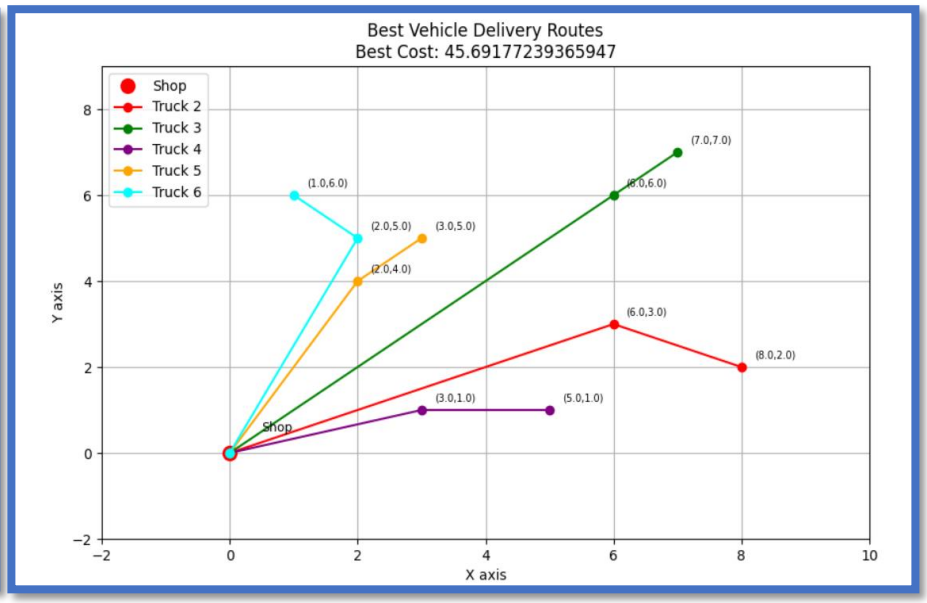
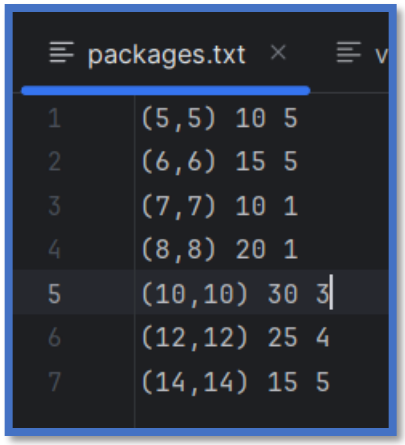


Figure 5: Test case A best solution paths

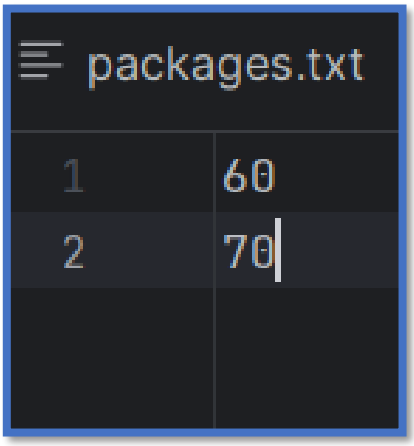
Test case -B-



packages.txt

```
(5,5) 10 5
(6,6) 15 5
(7,7) 10 1
(8,8) 20 1
(10,10) 30 3
(12,12) 25 4
(14,14) 15 5
```

Figure 8: Test case B packages file



packages.txt

```
60
70
```

Figure 7:Test case A vehicles file

Simulated Annealing output: -

```
Temp: 1.07 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.05 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.04 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.03 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.02 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.01 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 1.00 | Current Cost: 57.90 | Best Cost: 50.44
Temp: 0.99 | Current Cost: 57.90 | Best Cost: 50.44

=== Best Found Assignment ===
Vehicle 0 | Capacity: 60 | Packages: 3
   - Destination: (6.0, 6.0), Weight: 15.0, Priority: 5, ID: 1
   - Destination: (7.0, 7.0), Weight: 10.0, Priority: 1, ID: 2
   - Destination: (10.0, 10.0), Weight: 30.0, Priority: 3, ID: 4
Vehicle 1 | Capacity: 70 | Packages: 4
   - Destination: (5.0, 5.0), Weight: 10.0, Priority: 5, ID: 0
   - Destination: (8.0, 8.0), Weight: 20.0, Priority: 1, ID: 3
   - Destination: (12.0, 12.0), Weight: 25.0, Priority: 4, ID: 5
   - Destination: (14.0, 14.0), Weight: 15.0, Priority: 5, ID: 6

Actual Cost (without penalty): 33.94
Total Cost (with penalty): 50.44
```

Figure 10: Test case B best solution

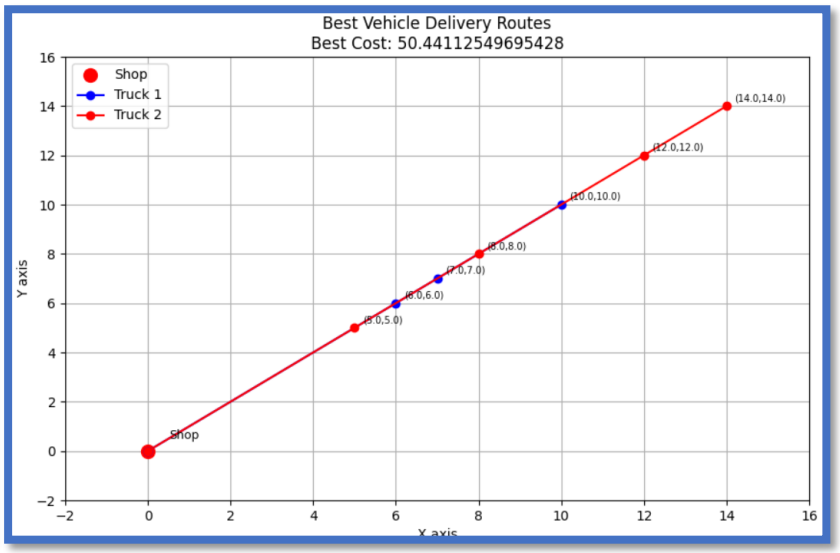


Figure 9: Test case B best solution paths

Genetic output: -

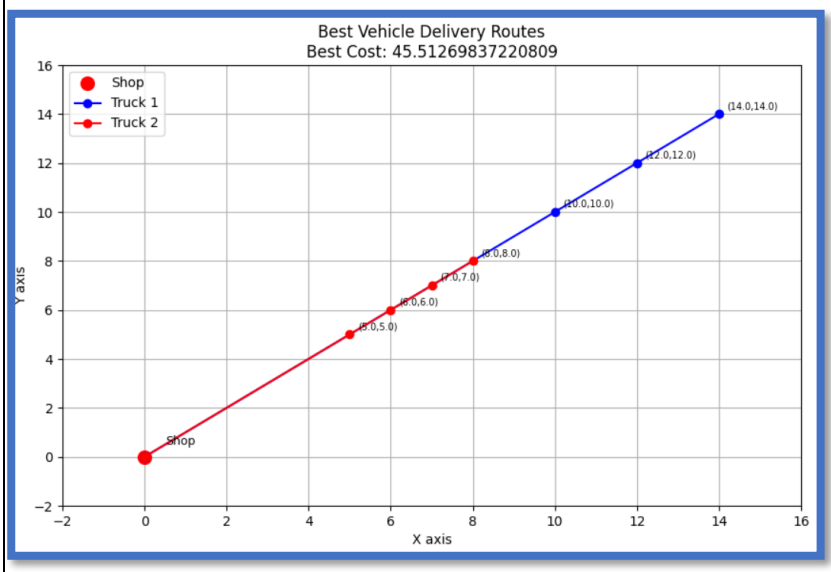


Figure 11:Test case B best solution Path

```
Generation 493, Best Fitness: -45.51
Generation 494, Best Fitness: -45.51
Generation 495, Best Fitness: -45.51
Generation 496, Best Fitness: -45.51
Generation 497, Best Fitness: -45.51
Generation 498, Best Fitness: -45.51
Generation 499, Best Fitness: -45.51

Actual Cost (without penalty): 31.11
Total Cost (with penalty): 45.51
Best Total Distance (Cost): 45.51
Vehicle 1 | Capacity: 70 | Packages: 3
  - Destination: (10.0, 10.0), Weight: 30.0, Priority: 3, ID: 4
  - Destination: (12.0, 12.0), Weight: 25.0, Priority: 4, ID: 5
  - Destination: (14.0, 14.0), Weight: 15.0, Priority: 5, ID: 6
Vehicle 1 | Capacity: 70 | Packages: 4
  - Destination: (5.0, 5.0), Weight: 10.0, Priority: 5, ID: 0
  - Destination: (6.0, 6.0), Weight: 15.0, Priority: 5, ID: 1
  - Destination: (7.0, 7.0), Weight: 10.0, Priority: 1, ID: 2
  - Destination: (8.0, 8.0), Weight: 20.0, Priority: 1, ID: 3
```

Figure 12: Test case B best solution